

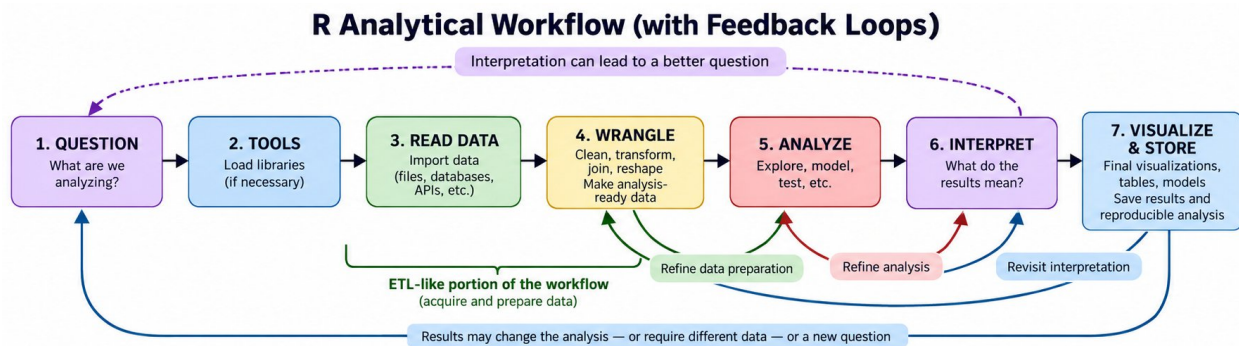
thuRsdays R Workshop

Lecture 1b: Data – They are not just tables anymore

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R Workflow:

Workflow means a set of steps that we use to perform analysis using R. Typically we use the following pipeline



- 1. Ask the question you want to answer and how to answer it**
 - a. This is important step outside of R
 - b. This is based on your domain
- 2. Identify tools to address the question**
 - a. Can R solve the problem.
 - b. Do we need libraries (additional functionality) to perform our analysis
 - c. Collect the required data for analysis
 - d. Read the data into R
 - e. Wrangle the data
 - i. Make the data analysis ready
 - ii. Deal with missing values, transformations
- 3. Analyze the data**
 - a. exploratory and confirmatory, inference, prediction, forecasting
- 4. Interpret the results**
 - a. Use preliminary visualization and summarizations
- 5. Refine the analysis**
 - a. May require going back to previous steps and repeat the analysis with a new set of insights and assumptions
- 6. Finalize the analysis and store**
 - a. Create final visualizations, summaries
 - b. Document your analysis script appropriately
 - c. Store data used, and R script used (reproducible analysis)